

H. Alternative Scaling Laws

H.1. Going Beyond Standard Power Laws

While we have found that in-context learning exhibits power law scaling with the number of demonstrations, the precise functional forms describing this phenomena have yet to be fully elucidated. We find some evidence for a non-standard scaling law which may offer a more precise description for in-context learning tasks. In particular, a Bounded Power Law scaling

$$nll(n) = C \left(1 + \frac{n}{n_c}\right)^{-\alpha} + K, \quad (6)$$

where C , α , n_c , and K are positive fit-parameters, captures the log-probabilities assigned by an LLM to its completions as a function of the number n of in-context demonstrations provided to the model rather well. The bounded power law has the additional advantage that it asymptotes to constant values for both limits $n \rightarrow 0$ and $n \rightarrow \infty$.

For example, we have evaluated a set of models from the same family⁵ but with different sizes on the model-generated psychopathy eval. The bounded power law provides an accurate description even for small n , as shown in Figure 13.

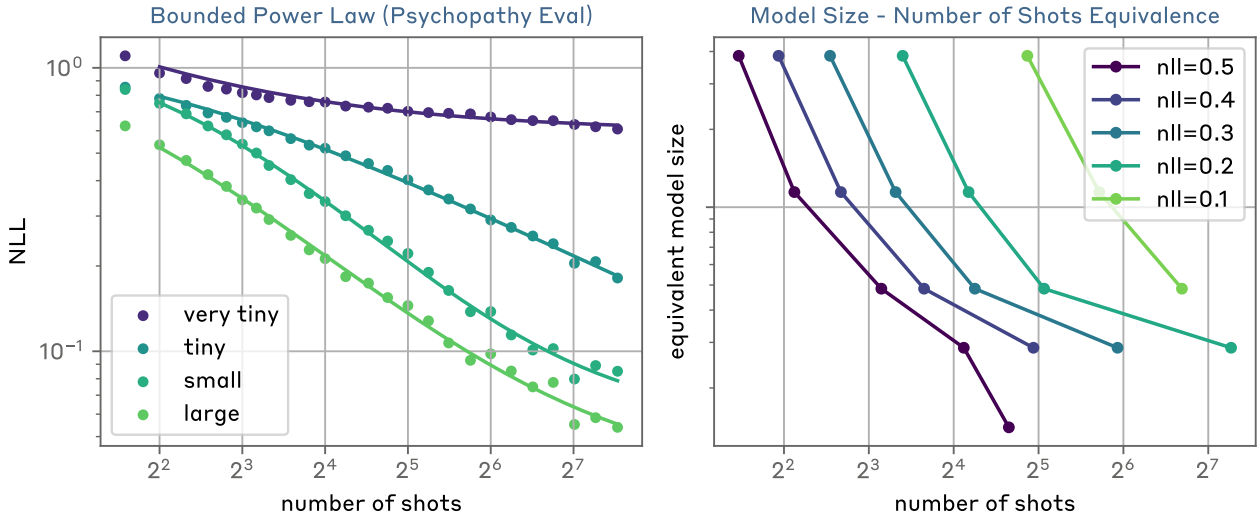


Figure 13. Bounded power law provides a more precise description for in-context learning tasks (left): We evaluated a set of models of different sizes on the psychopathy category of the Malevolent personality traits dataset. The bounded power law (Equation 6) captures the log-probabilities assigned by models to its completions as a function of the number of in-context demonstrations provided to the model well. **Equivalence between model size and the number of few-shot prompts (right):** For the same dataset, the few-shot/model-size equivalence plot demonstrates how, for a constant value of negative log-likelihood, variations in model size necessitate proportional changes in the number of few-shot demonstrations to maintain performance.

H.2. Few-Shot/Model-Size Equivalence Plots

We also leverage the bounded power law to generate a few-shot/model-size equivalence plots in Figure 13, elucidating the interplay between model scale N and the number n of few-shot demonstrations for a fixed negative log-likelihood nll . In other words, these plots delineate how, for a constant nll , variations in N necessitate proportional changes in n to maintain performance. By exploiting the bounded power law scaling, we can gain further insights into the intrinsic tradeoffs between model scale and number of demonstrations in many-shot learning.

H.3. Double Scaling Laws

It’s possible that the bounded power law (6) works well simply because it has an extra parameter. However, the effectiveness of the functional form (6) becomes clearer for more traditional datasets such as TruthfulQA (Lin et al., 2022) and GSM8K (Cobbe et al., 2021). Here we find some evidence that a double scaling law offers a more precise description of the scaling

⁵This family of models belong to an older lineage than the one Claude 2.0 belongs to.

trend, at least on some datasets. This is a generalization of the modified law (6) that captures the log-probabilities assigned by an LLM to its completions as a joint function of two key variables - the number n of in-context demonstrations provided to the model, as well as the overall model capacity as measured by parameter count N of the LLM:

$$nll(n, N) = C_n \left(1 + \frac{n}{n_c}\right)^{-\alpha_n} + C_N \left(1 + \frac{N}{N_c}\right)^{-\alpha_N}, \quad (7)$$

where C_n , C_N , α_n , α_N , n_c , and N_c are positive fit-parameters. The first term in the RHS of equation (7) is exactly the same as (6). The second term is the intercept term K of (6), but now a function of the size of the model. The main advantage of the above functional form is that the few-shot exponent α_n is completely independent of the size of the model, whereas the intercept is determined by the size of the model. As shown in Figures 14 and 15, we find that the above scaling law with 6-parameters provides an accurate fit for the TruthfulQA and GSM8K datasets for the same set of models as before with sizes ranging up to 52 Billion parameters.

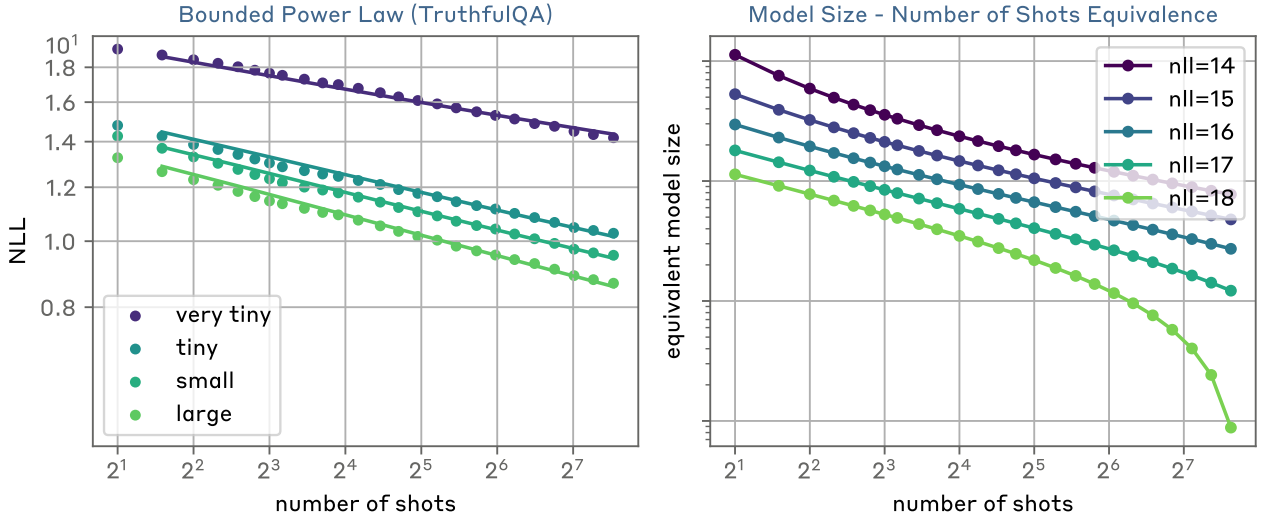


Figure 14. The double scaling law offers a precise description for the TruthfulQA dataset (left): We evaluated a set of models of different sizes on the TruthfulQA dataset. The double scaling law (7) accurately captures the log-probabilities assigned by models to completions as a joint function of two variables - the number of in-context demonstrations provided to the model, as well as the overall model capacity as measured by parameter count. **Equivalence between model size and the number of few-shot prompts (right):** We leveraged the double scaling law to generate a few-shot/model-size equivalence plot for the TruthfulQA dataset. This plot demonstrates how, for a constant negative log-likelihood, variations in model size necessitate proportional changes in the number of few-shot demonstrations to maintain performance.

The double scaling law (7) also offers valuable perspective into the efficacy of in-context learning. Specifically, we leverage the double scaling law to generate more accurate few-shot/model-size equivalence plots in Figures 14 and 15, elucidating the interplay between model scale N and the number n of few-shot demonstrations for a fixed negative log-likelihood nll . While power laws provide a useful approximate description, the double scaling law posits a more intricate relationship between model scale, data scale, and few-shot generalization capability. However, The double-scaling law has its limitations. For example, it doesn't seem to hold for the model generated psychopathy eval dataset. Further investigation into the exact form and universality of this scaling law across models and tasks offers rich potential for better understanding the drivers of in-context learning performance.

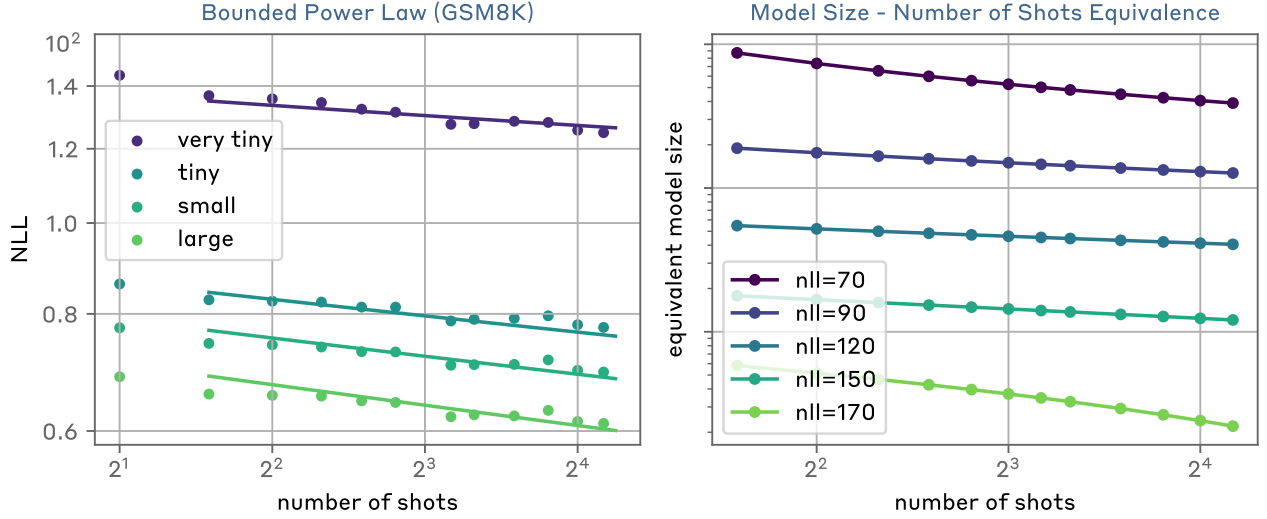


Figure 15. **The double scaling law also offers a precise description for the GSM8K dataset (left):** We evaluated the same set of models on the GSM8K dataset. The double scaling law (7) accurately captures the log-probabilities assigned by models to completions as a joint function of two variables - the number of in-context demonstrations provided to the model, as well as the overall model capacity as measured by parameter count. **Equivalence between model size and the number of few-shot prompts (right):** We leveraged the double scaling law to generate a few-shot/model-size equivalence plot for the GSM8K dataset. This plot demonstrates how, for a constant negative log-likelihood, variations in model size necessitate proportional changes in the number of few-shot demonstrations to maintain performance.

I. Prompt-based Defense Results

We compared the effectiveness of two prompt-modification based defenses against MSJ: In-Context Defense (ICD) by Wei et al. (2023b) and Cautionary Warning Defense (CWD). ICD prepends the incoming prompt with in-context demonstrations of desirable refusals. CWD is similar defense wherein the incoming prompt is both prepended and appended with a natural language string that cautions the model against being jailbroken.

To compare the effectiveness of these mitigations, we picked 256 harmful test questions across all of the harm categories in the malicious use-cases dataset. We then sampled MSJ strings of lengths varying from 8 to 205 and computed the attack success rate when the model is defended with ICD and CWD. We used the same refusal classification methodology we used in our experiments in Section 3.1.

The results can be seen in Figure 16. ICD leads to a minor reduction in attack success rate across all MSJ lengths we tested, while CWD consistently keeps the attack success rate down.

Below are additional details on the experimental conditions.

I.1. In-Context Defense Details

For ICD we selected a separate set of 20 additional harmful questions, recorded refusals (verified manually) from Claude 2.0, and then pre-pended the question-refusal pairs to the MSJ prompts. We used the same 20 questions for all of the ICD results. As Figure 16, ICD is only mildly effective in mitigating the effect of MSJ as the number of shots scale up. Although it’s possible that increasing the number of ICD shots accordingly could act as a counter-measure, we observe that it would be impractical to have a defense mechanism that needs to be adjusted based on the scale of the attack.

I.2. Cautionary Warning Defense Details

We modified the prompt in two ways. First, we pre-pended a cautionary paragraph, warning Claude 2.0 that it was about to receive a question and that it might trick it into offering an answer that violated its principles. Second, we reinforced the warning by appending a similar string at the end of the user prompt. The complete prompt would therefore look like:

warning $\rightarrow$ n-shot msj $\rightarrow$ question $\rightarrow$ warning